Wk 5 Supervision Problems 2

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Convergence Rates

Solve the assignment problems from this old analysis workbook.

An excellent method for calculating square roots is the Newton-Raphson method which you may have met at A-level. When applied to the problem of calculating $\sqrt{2}$ this leads to the sequence given by: $a_0 = 2$ and $a_{n+1} = \frac{1}{a_n} + \frac{a_n}{2}$.

Exercise 8 Use a calculator to calculate a_1, a_2, a_3, a_4. Compare them with $\sqrt{2}$.

Assignment 16
Use induction to show that $1 \leq a_n \leq 2$ for all n. Assuming that (a_n) converges, show that the limit must be $\sqrt{2}$.

We will now show that the sequence converges to $\sqrt{2}$ like a bat out of hell.

Assignment 17
Show that $(a_{n+1} - \sqrt{2}) = \frac{(a_n - \sqrt{2})^2}{2a_n}$. Using this identity show by induction that $|a_n - \sqrt{2}| \leq \frac{1}{2^{2^n}}$. How many iterations do you need before you can guarentee to calculate $\sqrt{2}$ to within an error of 10^{-100} (approximately 100 decimal places)?

Quick history lesson: In my year's analysis I module, we had workbooks we had to submit which contained ***over 15 compulsory assignment problems every week!***

Induction

Here is an induction proof that all horses are the same colour:

> Clearly, a group of one horse is a group in which all horses are the same colour. This constitutes the base case.
> Now suppose that any group of n horses are always the same colour, and consider a group of $n+1$ horses. We let this group of horses be $\{H_1, H_2, \ldots, H_{n+1}\}$ and consider removing just one horse. $\{H_2, H_3, \ldots, H_{n+1}\}$ is a group of n horses, which must all be the same colour, and $\{H_1, H_2, \ldots, H_n\}$ is a group of n horses, which must all be the same colour, so it follows that every horse in $\{H_1, H_2, \ldots, H_{n+1}\}$ are all the same colour. This completes the inductive step.
> We conclude by induction that in any group of horses of any size, all horses are the same colour.

Do you agree?